AE4ASM003 — Linear Modelling (incl. F.E.M.)

Homework Assignment 3 (Part 1)

AI assistance was used to format tables in LaTeX and improve presentation.

Author

Matilde Barros (6547788)

Date

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Faculty of Aerospace Engineering
Delft University of Technology

Introduction

In this part it's considered a prismatic cantilever beam of length L and rectangular cross-section $(w \times h)$, clamped at $x = 0$ and free at $x = L$. The x-axis runs along the beam (from clamp to free end) and the y-axis is transverse. Transverse deflection $v(x)$ is taken negative downward. The beam is subjected to a line load in the y-direction $q_y(x) = -(ax + b)$ and to a constant axial force $F > 0$ (tension). Linear elasticity and Euler–Bernoulli assumptions are used. Areas and inertias are $A = wh$ and $I_z = \frac{wh^3}{12}$.

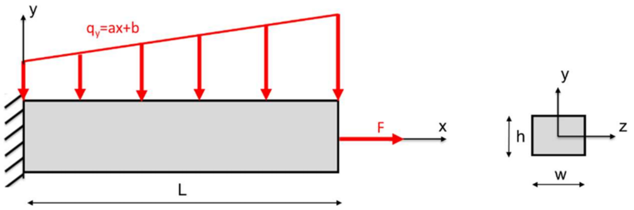

Figure 1: Cantilever, axes and loading directions.

Symbol	Value	Symbol	Value
E	70 GPa	F	1.5 kN
L	1200 mm	w	70 mm
b	5 N/mm	h	40 mm
a	0.001 N/mm^2		

Figure 2: Given numerical data.

$$I_z = \frac{wh^3}{12} = 3.733 \times 10^5 \text{ mm}^4, \qquad A = wh = 2800 \text{ mm}^2$$

Question a)

1-Deflection in y-direction of the midplane as a function of x

According to Euler–Bernoulli beam theory, the governing differential equation for the vertical deflection $v(x)$ (taken negative downward) under the transverse load distribution $q_y(x) = -(ax + b)$ is

$$EI_z\, v^{(4)}(x) = -(ax + b).$$

Integrating four times gives the general form

$$v(x) = -\Big(\frac{a}{120EI_z}x^5 + \frac{b}{24EI_z}x^4 + \frac{C_1}{6}x^3 + \frac{C_2}{2}x^2 + C_3x + C_4\Big).$$

Applying the cantilever boundary conditions

$$v(0) = 0, \quad v'(0) = 0, \quad v''(L) = 0, \quad v^{(3)}(L) = 0,$$

leads to $C_3 = C_4 = 0$, $C_1 = -\frac{a}{2EI_z}L^2 - \frac{b}{EI_z}L$, and $C_2 = \frac{a}{3EI_z}L^3 + \frac{b}{2EI_z}L^2$.

Thus, the vertical deflection of the mid-plane is

$$v(x) = \frac{-1}{EI_z}\left[\frac{a}{120}x^5 + \frac{b}{24}x^4 - \Big(\frac{aL^2}{12} + \frac{bL}{6}\Big)x^3 + \Big(\frac{aL^3}{6} + \frac{bL^2}{4}\Big)x^2\right], \quad 0 \le x \le L.$$

In particular, the tip deflection at $x = L$ is

$$v(L) = \frac{-1}{EI_z}\left(\frac{11}{120}aL^5 + \frac{1}{8}bL^4\right).$$

Substituting into the deflection expression gives

$$v(x) = \frac{-1}{2.613 \times 10^{10}}\left[8.33 \times 10^{-6}\, x^5 + 0.208\, x^4 - 1120\, x^3 + 2.088 \times 10^6\, x^2\right],$$

with x in mm, $v(x)$ in mm.

The tip deflection is then

$$v(L) \approx -58.3 \text{ mm}.$$

2-The maximum axial extension of the beam due to the axial force

We take $u(x)$ positive in the beam's axial (tension) direction. For an axially loaded member the normal stress is

$$\sigma = \frac{F}{A},$$

so by Hooke's law the strain is

$$\varepsilon = \frac{\sigma}{E} = \frac{F}{EA}.$$

Since strain is also defined as $\varepsilon = \frac{du}{dx}$, the axial displacement along the beam is

$$u(x) = \frac{F}{EA}x.$$

The displacement increases linearly from the clamped end ($x = 0$) to the free end ($x = L$), so the maximum axial extension is

$$u_{\max} = u(L) = \frac{F\,L}{E\,A}.$$

Substituting the values, we obtain

$$u_{\max} = \frac{1500 \times 1200}{(70{,}000) \times 2800} \text{ mm} = \frac{1.80 \times 10^6}{1.96 \times 10^8} \text{ mm} \approx 9.18 \times 10^{-3} \text{ mm}.$$

3-The maximum tensile stress in the beam

The total normal stress is the sum of axial and bending contributions:

$$\sigma(x, y) = \frac{F}{A} + \frac{M(x)\,y}{I_z}, \qquad |y| \le \tfrac{h}{2}.$$

From equilibrium we have

$$\frac{dV}{dx} = -(ax + b), \qquad \frac{dM}{dx} = -V(x).$$

Integrating and applying the free-end condition $V(L) = 0$ gives

$$V(x) = -\tfrac{a}{2}x^2 - bx + \tfrac{a}{2}L^2 + bL,$$

so the shear is maximum at the clamp $x = 0$, with

$$|V|_{\max} = \tfrac{a}{2}L^2 + bL.$$

Integrating again and applying $M(L) = 0$ gives

$$M(x) = \tfrac{a}{6}x^3 + \tfrac{b}{2}x^2 - \Big(\tfrac{a}{2}L^2 + bL\Big)x + \tfrac{a}{3}L^3 + \tfrac{b}{2}L^2.$$

The bending moment magnitude is therefore maximum at the clamp $x = 0$, with

$$M(0) = \tfrac{a}{3}L^3 + \tfrac{b}{2}L^2.$$

Thus the maximum tensile stress is

$$\sigma_{\max} = \frac{F}{A} + \left(\tfrac{a}{3}L^3 + \tfrac{b}{2}L^2\right)\frac{h}{2I_z}.$$

Substituting the values, we obtain

$$\sigma_{\max} = \frac{1500}{2800} + \Big(\tfrac{0.001}{3}\cdot 1200^3 + \tfrac{5}{2}\cdot 1200^2\Big)\frac{40}{2\cdot 373333} = 224.25 \text{ MPa},$$

which occurs at the clamped end ($x = 0$), at the outermost fiber ($y = h/2$).

Question b)

Because only a transverse linearly varying load is applied, the derivation is carried out first for a beam element and then embedded in the frame element.

The element axis is $x \in [0, L]$ from node i to node j. The nodal degrees of freedom are the transverse displacements u_{i1}, u_{j1} and the rotations (slopes) $u_{i2} = \frac{dv}{dx}|_{x=0}$ and $u_{j2} = \frac{dv}{dx}|_{x=L}$. Since the beam element has four dofs, a third–order polynomial is adopted for $v(x)$:

$$v(x) = c_1 + c_2x + c_3x^2 + c_4x^3.$$

With $x_i = 0$ and $x_j = L$, the boundary relations are

$$u_{i1} = v(0) = c_1, \qquad u_{i2} = v'(0) = c_2,$$

$$u_{j1} = v(L) = c_1 + c_2L + c_3L^2 + c_4L^3, \qquad u_{j2} = v'(L) = c_2 + 2c_3L + 3c_4L^2.$$

Solving for the coefficients in terms of the nodal degrees of freedom gives

$$c_1 = u_{i1}, \qquad c_2 = u_{i2},$$

$$c_3 = -\frac{3u_{i1}}{L^2} - \frac{u_{i2}}{L} + \frac{3u_{j1}}{L^2} - \frac{u_{j2}}{L}, \qquad c_4 = \frac{2u_{i1}}{L^3} + \frac{u_{i2}}{L^2} - \frac{2u_{j1}}{L^3} + \frac{u_{j2}}{L^2}.$$

Substituting back and rearranging in terms of the nodal degrees of freedom yields the Hermite shape functions

$$N_1 = \frac{L^3 - 3Lx^2 + 2x^3}{L^3}, \qquad N_2 = \frac{L^2x - 2Lx^2 + x^3}{L^2}, \qquad N_3 = \frac{3Lx^2 - 2x^3}{L^3}, \qquad N_4 = \frac{x^3 - Lx^2}{L^2},$$

and the interpolation

$$v(x) = N_1\, u_{i1} + N_2\, u_{i2} + N_3\, u_{j1} + N_4\, u_{j2}.$$

The external work of the distributed load over the element is

$$W_p = \int_0^L q_y(x)\, v(x)\, dx = \int_0^L q_y(x)\begin{pmatrix}N_1 & N_2 & N_3 & N_4\end{pmatrix}\begin{pmatrix}u_{i1}\\ u_{i2}\\ u_{j1}\\ u_{j2}\end{pmatrix} dx = \left(\int_0^L \begin{pmatrix}N_1 & N_2 & N_3 & N_4\end{pmatrix} q_y(x)\, dx\right)(u).$$

.

The consistent nodal loads follow from work equivalence:

$$\mathbf{F} = \begin{pmatrix} F_{i1} \\ F_{i2} \\ F_{j1} \\ F_{j2} \end{pmatrix} = \frac{\partial W_p}{\partial (u_{i1}, u_{i2}, u_{j1}, u_{j2})^T} = \int_0^L \begin{pmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \end{pmatrix} q_y(x)\, dx.$$

The four integrals for $q_y(x) = -(ax+b)$ are evaluated explicitly.
Integral for N_1:

$$\begin{aligned} \int_0^L N_1 q_y \, dx &= -\int_0^L \frac{L^3 - 3Lx^2 + 2x^3}{L^3}\,(ax+b)\,dx \\ &= -\frac{a}{L^3}\int_0^L (L^3 x - 3Lx^3 + 2x^4)\,dx \;-\; \frac{b}{L^3}\int_0^L (L^3 - 3Lx^2 + 2x^3)\,dx \\ &= -\frac{L}{20}\big(3aL + 10b\big). \end{aligned}$$

Integral for N_2:

$$\begin{aligned} \int_0^L N_2 q_y \, dx &= -\int_0^L \frac{L^2 x - 2Lx^2 + x^3}{L^2}\,(ax+b)\,dx \\ &= -\frac{a}{L^2}\int_0^L (L^2 x^2 - 2Lx^3 + x^4)\,dx \;-\; \frac{b}{L^2}\int_0^L (L^2 x - 2Lx^2 + x^3)\,dx \\ &= -\frac{L^3}{30}a\frac{L^2}{12}b. \end{aligned}$$

Integral for N_3:

$$\begin{aligned} \int_0^L N_3 q_y \, dx &= -\int_0^L \frac{3Lx^2 - 2x^3}{L^3}\,(ax+b)\,dx \\ &= -\frac{a}{L^3}\int_0^L (3Lx^3 - 2x^4)\,dx \;-\; \frac{b}{L^3}\int_0^L (3Lx^2 - 2x^3)\,dx \\ &= -\frac{L}{20}\big(7aL + 10b\big). \end{aligned}$$

Integral for N_4:

$$\begin{aligned} \int_0^L N_4 q_y \, dx &= -\int_0^L \frac{x^3 - Lx^2}{L^2}\,(ax+b)\,dx \\ &= -\frac{a}{L^2}\int_0^L (x^4 - Lx^3)\,dx \;-\; \frac{b}{L^2}\int_0^L (x^3 - Lx^2)\,dx \\ &= \frac{L^3}{20}a + \frac{L^2}{12}b. \end{aligned}$$

Collecting the results (beam, local order $\{u_{i1}, u_{i2}, u_{j1}, u_{j2}\}$):

$$\mathbf{F} = \begin{pmatrix} -\frac{3}{20}aL^2 - \frac{1}{2}b\,L \\ -\frac{1}{30}aL^3 - \frac{1}{12}b\,L^2 \\ -\frac{7}{20}aL^2 - \frac{1}{2}b\,L \\ \frac{1}{20}aL^3 + \frac{1}{12}b\,L^2 \end{pmatrix}.$$

For the uniform-load check ($a = 0$, $b = w$),

$$\{F_{i1}, F_{i2}, F_{j1}, F_{j2}\} = \left\{ -\frac{wL}{2},\ -\frac{wL^2}{12},\ -\frac{wL}{2},\ +\frac{wL^2}{12} \right\}.$$

For the 2D frame, the local DOF order is $\{u_{i1}, u_{i2}, u_{i3}, u_{j1}, u_{j2}, u_{j3}\} = \{u_x, v, \theta, u_x, v, \theta\}$. Because the load is purely transverse, the axial components are zero, and the beam result occupies the v–θ slots:

$$\mathbf{F}^{\,\text{frame}} = \begin{pmatrix} 0 \\ -\frac{3}{20}aL^2 - \frac{1}{2}b\,L \\ -\frac{1}{30}aL^3 - \frac{1}{12}b\,L^2 \\ 0 \\ -\frac{7}{20}aL^2 - \frac{1}{2}b\,L \\ \frac{1}{20}aL^3 + \frac{1}{12}b\,L^2 \end{pmatrix}.$$

Question c) and d)

It's considered the cantilever beam with a downward distributed load $q(x) = -(ax + b)$ and a tip axial force F at $x = L$. Analytical benchmarks are

$$u(L) = \frac{FL}{EA}, \qquad v(L) = -\frac{1}{EI_z}\left(\frac{11}{120}\,aL^5 + \frac{1}{8}\,bL^4\right), \qquad \theta(L) = -\frac{1}{EI_z}\left(\frac{aL^4}{8} + \frac{bL^3}{6}\right).$$

With a consistent distributed-load vector and cantilever boundary conditions, the finite element (FE) solution reproduces these tip values to machine precision for every tested mesh size N_e. The free-end resultants are enforced exactly, so the tip degrees of freedom follow from equilibrium. A tip check alone is not informative for mesh adequacy, since interior shape error may still be present. The study therefore evaluates the full deflection field and a local stress at the critical section.

The FE–analytical overlay in Fig. 3 shows that all meshes satisfy the clamp and meet the exact tip, remaining discrepancies are interior shape errors due to approximating a quartic exact deflection with piecewise-cubic Hermite functions. For $N_e = 2$ and $N_e = 3$ the curves track the analytical solution closely, but small separations are still visible under magnification. These discrepancies shrink monotonically with refinement and are visually negligible by $N_e = 4$.

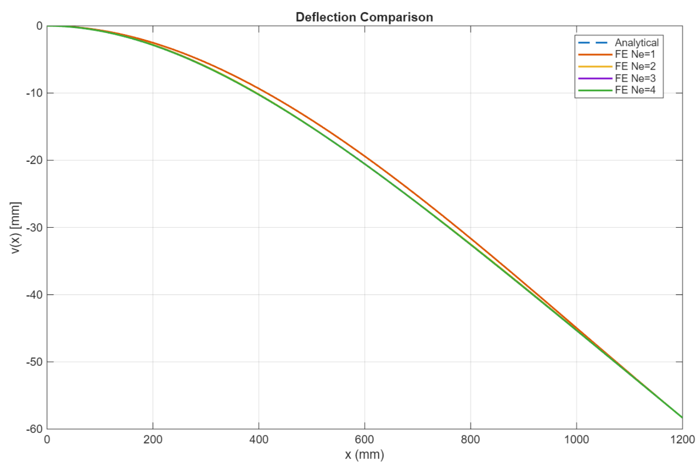

Figure 3: Deflection $v(x)$ [mm]: FE vs analytical.

Field accuracy is quantified with the one-dimensional L^2 norm of the deflection along the span,

$$\|v - v_h\|_{L^2(0,L)} = \left(\int_0^L \frac{1}{L} [v(x) - v_h(x)]^2 \, dx \right)^{1/2},$$

including only v because the axial field $u(x) = \frac{Fx}{EA}$ is linear and is reproduced exactly by the axial interpolation. For cubic Hermite interpolation (polynomial degree $p = 3$) the expected behavior is $\|v - v_h\|_{L^2} = O(h^{p+1}) = O(h^4)$; this slope is observed in the data, with roughly a 16× error reduction when h is halved.

Table 1: L^2 error of the deflection vs. number of elements.

N_e	$\|v - v_h\|_{L^2(0,L)}$ [mm]
1	7.376×10^{-1}
2	4.617×10^{-2}
3	9.122×10^{-3}
4	2.887×10^{-3}
5	1.182×10^{-3}
10	7.390×10^{-5}
20	4.619×10^{-6}

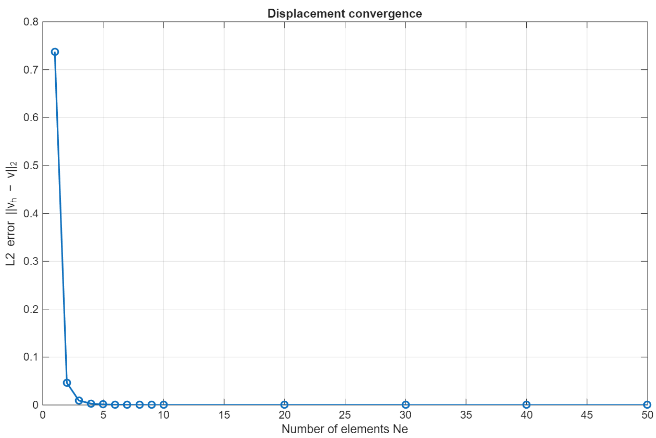

Figure 4: Convergence of the deflection in the L^2 norm.

As a local design-relevant quantity at the critical section, the normal stress at the fixed end is tracked and evaluated at the outer fiber,

$$\sigma(0, h/2) = \frac{F}{A} + \frac{M(0)\,h}{2I_z}.$$

For $q(x) = -(ax + b)$ the exact curvature is quadratic in x, whereas the FE curvature is piecewise linear, so the stress converges more slowly, about $O(h^2)$. This behavior appears in Fig. 5 and Table 2: the relative error drops by $\sim 4\times$ for each mesh halving; the 1% line is crossed at $N_e = 4$.

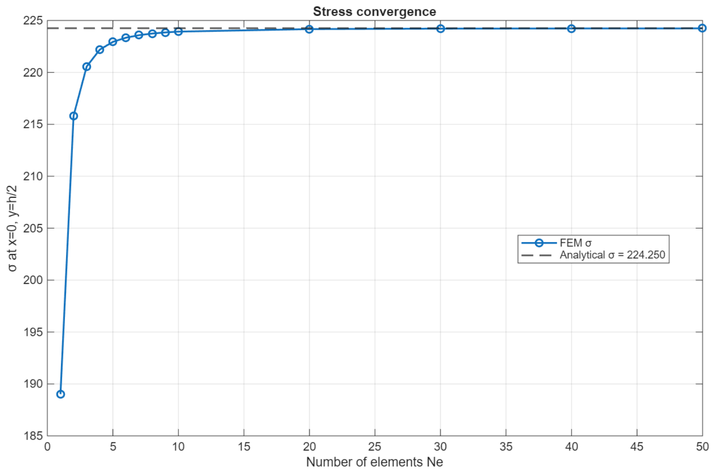

Figure 5: Stress convergence at the clamp ($x = 0,\ y = +h/2$).

Table 2: Relative error of the clamp stress at the outer fiber ($x = 0,\ y = +h/2$).

N_e	$\dfrac{\lvert\sigma_h(0,h/2) - \sigma_{\mathrm{ref}}\rvert}{\sigma_{\mathrm{ref}}}$ [%]
1	15.71
2	3.755
3	1.644
4	0.917
5	0.584
10	0.145
20	0.036

Mesh adequacy is decided by combining a global and a local tolerance and accepting the solution when

$$\|v - v_h\|_{L^2(0,L)} < 3 \times 10^{-3}\ \mathrm{mm} \qquad \text{and} \qquad \frac{\lvert\sigma_h(0,h/2) - \sigma_{\mathrm{ref}}\rvert}{\sigma_{\mathrm{ref}}} < 1\%.$$

The first condition controls whole-span accuracy of the serviceability quantity (deflection), and the second safeguards the design hotspot at the clamp where stresses peak and convergence is slower. Both criteria are first satisfied at $N_e = 4$; at this mesh the RMS deflection error is 2.887×10^{-3} mm and the clamp-stress error is 0.917%. Finer meshes further reduce already small errors without changing the physical conclusions.

Mesh convergence balances accuracy and efficiency. After an initial result, the chosen discretisation (number of nodes and elements) must be checked for adequacy. Increasing N_e indefinitely improves accuracy but raises computational cost; the practical approach is to monitor error under uniform refinement and terminate when prescribed tolerances, such as the dual criteria above, are met. In this example, the smallest mesh that satisfies both accuracy and efficiency is $N_e = 4$.

Question e)

If the cross-section varies with constant height h and a linearly varying width $w(x)$, the geometric properties become

$$A(x) = h\,w(x), \qquad I_z(x) = \frac{h^3\,w(x)}{12},$$

so both $A(x)$ and $I(x)$ are linear functions of x. The question is whether replacing these by their value at the element midpoint provides an accurate element formulation.

For the axial contribution, the element stiffness reads

$$k_{\text{ax}} = \int_0^\ell E\,A(x)\,N_u'(x)^T N_u'(x)\,dx.$$

In a linear axial element, $N_u'(x)$ is constant, so the integrand is proportional to $A(x)$, i.e. a linear polynomial in x. A midpoint evaluation (equivalent to 1-point Gauss quadrature) integrates this exactly. Thus, using the mid-element cross-section is exact for the axial part when $w(x)$ varies linearly.

For the bending contribution, the stiffness is

$$k_{\text{bend}} = \int_0^\ell E\,I_z(x)\,B(x)^T B(x)\,dx,$$

where $B(x)$ involves second derivatives of the Hermite shape functions. These derivatives are linear in the local coordinate, so $B^T B$ is quadratic. Multiplying by $I_z(x)$ gives a cubic integrand. A midpoint evaluation, exact only for polynomials up to degree one, cannot integrate this cubic function exactly. Approximating $I_z(x)$ by its mid-element value effectively treats the section as piecewise constant, yielding only a first-order approximation of the bending stiffness. This approach converges with mesh refinement, but that improvement comes from using more elements, not from the midpoint assumption itself. It requires a finer mesh to achieve the same accuracy as proper integration.

Therefore, using the mid-element cross-section is exact for the axial stiffness under a linear width variation, but it is not a good approximation for the bending stiffness unless the change of width within the element is very small.

Question f)

Consider a uniform truss member with constant EA subjected to a linearly varying axial load $p(x)$. The governing equation $(EA\,u')' + p(x) = 0$ implies an exact displacement $u(x)$ that is cubic in x, so the exact axial strain $u'(x)$ and force $N(x) = EA\,u'(x)$ are quadratic and vary smoothly along the member.

Using a single quadratic truss element over the span introduces a curved displacement field across the whole member and a linearly varying strain within the element. This does not reproduce the exact quadratic strain, but it follows the smooth variation much more closely than a piecewise-constant strain. For the same total number of degrees of freedom, a quadratic interpolation avoids artificial slope jumps at interior points, yields smaller energy and stress-resultant errors, and achieves a higher convergence rate for smooth solutions.

Modelling the same member with multiple linear truss elements produces a displacement that is only piecewise linear and a strain that is constant in each element with jumps at the inter-element nodes. Accuracy then relies on mesh refinement: the response is approximated in segments, and more elements are needed to mimic the global curvature that a single quadratic can represent. This piecewise-constant strain is a poorer surrogate for the true quadratic distribution when the member and loading are smooth.

Under the stated conditions, uniform properties, no internal discontinuities, and a smoothly varying axial load, the preferred choice is a single quadratic truss element across the member because it represents the curved displacement and smoothly varying strain more faithfully for the same modelling effort.

References

[1] D. Peeters, *Linear Modelling (incl. FEM)* , TU Delft, Version 4, 2025.

AE4ASM003 — LINEAR MODELLING (INCL. F.E.M.)

Homework Assignment 3 (Part 2)

AI assistance was used to format tables in LaTeX and improve presentation.

Author

Matilde BARROS (6547788)

Date

September 29, 2025

Faculty of Aerospace Engineering

Delft University of Technology

Introduction

This part idealises the covered–walkway segment as a planar frame with a vertical member AB clamped at A and a quarter-circular member BC terminating at the free tip C. A vertical line load of magnitude $q = 20$ N/mm acts along BC and a concentrated tip load $F = 500$ N is applied at C.

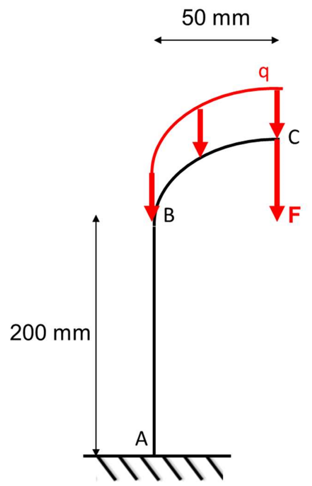

Figure 1: Geometry, boundary conditions and loading.

Question a)

The structure was modelled in Abaqus using B23 beam elements and the boundary conditions and loads specified in the assignment. The corresponding `.cae` file is delivered alongside this report.

Question b)

Euler–Bernoulli beam theory enforces plane sections to remain normal to the deformed axis (no transverse shear), which is accurate for slender members. Timoshenko theory keeps sections plane but allows rotation independent of the centerline slope, so transverse shear is included, this is required when the span is short relative to the section depth.

Element names: B22 = beam, 2D, quadratic bending interpolation (used here as a Timoshenko element) and B23 = beam, 2D, cubic bending interpolation (Euler–Bernoulli).

With $h = 20\,\text{mm}$,

$$\frac{L_{AB}}{h} = \frac{200}{20} = 10, \qquad \frac{L_{BC}}{h} = \frac{(\pi/2)\,50}{20} \approx 3.9.$$

Typical Euler–Bernoulli guidelines require $L/h \gtrsim 15$–20. Hence AB is borderline slender, while the $90°$ arc BC is stout, under the vertical line load on BC and the point load at C, shear deformation on BC is not negligible. Enforcing no transverse shear deformation (B23) would over-stiffen BC and underestimate the response at C.

Accordingly, the appropriate element for both members is a quadratic Timoshenko beam (B22): sufficiently accurate on AB at $L/h = 10$ and necessary on BC at $L/h \approx 4$ to capture transverse shear. A small number of quadratic elements along the $90°$ arc with local frames aligned to the tangent preserves the correct in-plane bending axis without unnecessary refinement.

Question c)

The finite element formulation, the loads and the boundary conditions are kept fixed and only the mesh is varied. Local beam axes follow the geometry and the distributed load on BC is applied as a consistent line load on the curved reference line. The analysed sequence of meshes is

$$(1,2) \to (2,4) \to (3,6) \to (4,8) \to (5,10) \to (6,12),$$

and an overkill model with three hundred elements on AB and six hundred elements on BC is computed to reveal response features and to provide a numerical reference. In this notation (N_{AB}, N_{BC}) means that N_{AB} B23 beam elements are placed along member AB and that N_{BC} B23 beam elements are placed along member BC. For example the pair $(1,2)$ uses one element on AB and two elements on BC, and the pair $(3,6)$ uses three elements on AB and six elements on BC.

A different number of elements is used on each member in order to reduce the total number of elements while keeping accuracy where it is needed. Member BC is curved and carries the distributed line load. The bending field varies more rapidly along BC than along the straight member AB. Shorter elements on BC improve the interpolation of curvature and improve the numerical integration of the line load for the same overall cost. Member AB still requires resolution near the clamp where the bending gradient is steep. A small number of elements on AB is therefore sufficient to resolve the root region without spending elements where they bring little improvement.

The overall behaviour is that of a bending dominated cantilever. The vertical line load on BC and the tip force at C generate shear and bending along the arc. These resultants pass through the knee at B and build up toward the clamp at A. The internal moment increases from the tip to the base. The rotation grows monotonically from zero at the clamp to its maximum at the tip. The transverse displacement varies smoothly along the member. The normal stress field peaks at the clamp with the expected tension and compression pattern across the section. Curvature remains continuous across B and this produces smooth transfer of stress and moment at the knee. The support reactions recover the applied resultants and this confirms equilibrium and consistent boundary conditions.

This were the monitored quantities:

- v_C vertical displacement at the tip. This represents the serviceability deflection at the design point. It converges quickly because it is a displacement quantity and it reflects the global bending response.

- θ_C in plane rotation at the tip. Rotation governs serviceability and equals the line integral of curvature along the load path. It is more sensitive than displacement but it stabilises early and it complements the displacement at the tip.

- $R_{y,A}$ vertical reaction at the clamp. This verifies global equilibrium and correct load application and boundary conditions. It is weakly sensitive to mesh refinement and it provides an anchor for the solution.

- M_A reaction moment at the clamp. This quantifies the design relevant bending demand where stresses peak. It checks consistency between reactions and the internal section moment near the support and it indicates whether the root gradient is sufficiently resolved.

- M_B internal section moment at the knee. This assesses smooth transfer of bending through the geometric transition. It depends on curvature which is a second derivative and it is therefore the slowest quantity to converge. It is used as a secondary acceptance metric in addition to the tip and base checks so that the mesh is only accepted when the knee moment is within an agreed tolerance and shows monotonic stabilisation with refinement.

Table 1: Mesh study with raw results from Abaqus

(N_{AB}, N_{BC})	$R_{y,A}$ [N]	M_A (RM3) [N mm]	M_B (SM1) [N mm]	v_C [mm]	θ_C [rad]
(1,2)	2030.73	55342.7	54875.7	-1.37846	-2.77142×10^{-2}
(2,4)	2061.11	53935.9	55663.5	-1.39670	-2.77890×10^{-2}
(3,6)	2070.26	53548.9	54537.2	-1.37499	-2.74629×10^{-2}
(4,8)	2070.57	53543.2	54097.2	-1.37044	-2.74031×10^{-2}
(5,10)	2070.68	53541.4	53893.2	-1.36884	-2.73820×10^{-2}
(6,12)	2070.73	53540.7	53782.5	-1.36813	-2.73726×10^{-2}
(300,600)	2070.80	53539.8	53539.8	-1.36358	-2.73599×10^{-2}

Convergence by successive changes is evaluated with

$$\Delta q^{(k)} = \frac{\left|q^{(k)} - q^{(k-1)}\right|}{\left|q^{(k)}\right|} \times 100\%.$$

The acceptance uses only the successive change method and follows the physics of the problem.

Table 2: Successive changes expressed in percent

from to	$\Delta R_{y,A}$	ΔM_A	ΔM_B	Δv_C	$\Delta \theta_C$
(1,2) to (2,4)	1.47	2.61	1.42	1.31	0.27
(2,4) to (3,6)	0.44	0.72	2.07	1.58	1.19
(3,6) to (4,8)	0.015	0.011	0.81	0.33	0.22
(4,8) to (5,10)	0.005	0.003	0.38	0.12	0.08
(5,10) to (6,12)	0.002	0.001	0.21	0.05	0.03
(6,12) to (300,600)	0.003	0.17	0.45	0.33	0.046

From $(2,4) \to (3,6)$ the changes are $\Delta v_C = 1.58\%$, $\Delta\theta_C = 1.19\%$, $\Delta R_{y,A} = 0.44\%$, $\Delta M_A = 0.72\%$ and $\Delta M_B = 2.07\%$ and from $(3,6) \to (4,8)$ they drop to $\Delta v_C = 0.33\%$, $\Delta\theta_C = 0.22\%$, $\Delta R_{y,A} = 0.015\%$, $\Delta M_A = 0.011\%$ and $\Delta M_B = 0.81\%$. The step $(3,6) \to (4,8)$ therefore yields much smaller changes than the earlier steps, which shows that $(3,6)$ already lies on the flat part of the convergence curve.

In conclusion the mesh with **three elements on AB and six elements on BC (3,6)**, represented in Figure 2, is the best compromise between accuracy and computational efficiency, any further refinement produces only marginal gains at a higher cost. It delivers stable tip displacement and rotation. It delivers stable support reaction and base moment. It keeps the knee moment under control and trending down. It uses only nine elements rather than a larger uniform mesh and it achieves the same design decisions.

Even better accuracy per element could be obtained by biasing the mesh within each member with smaller elements very close to the clamp and in the region where curvature is highest. This option is not used here in order to keep the refinement pattern simple and to show clearly how each global refinement step affects the monitored quantities.

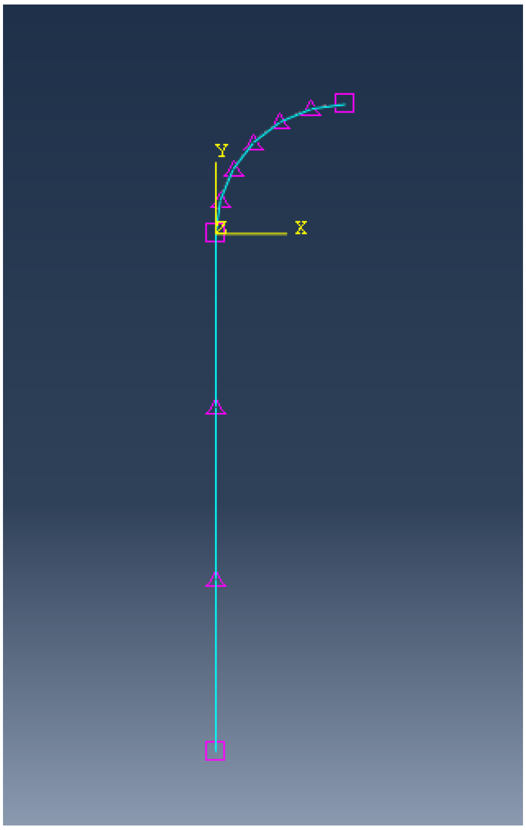

Figure 2: Adopted mesh with three elements on AB and six elements on BC (3,6).

Question d) and e)

To determine which cross-section results in the smallest mass, the mass of each shape was estimated from its geometry. Since the material density ρ is known, the total mass follows directly from the product of density and volume:

$$M = \rho V = \rho A l$$

where M is the mass, A the cross-sectional area, and l the member length. For a fixed material and length, only the cross-sectional area influences the mass. Therefore, comparing the areas of the different shapes is enough to establish their relative weights.

Table 3: Cross-sectional area of the considered shapes.

Shape	Area [mm^2]
I-profile, $t = 2$ mm	52.0
I-profile, $t = 1.5$ mm	40.5
Hollow rectangular profile	64.0

From Table 3, the I-profile with $t = 1.5$ mm is the lightest option, followed by the thicker I-profile and finally the hollow rectangular section, which has the largest area (and therefore the highest mass).

The next step was to evaluate the bending stiffness of each shape through its second moment of area I, presented in Table 4.

Table 4: Second moment of area of the considered shapes.

Shape	Moment of Inertia [mm^4]
I-profile, $t = 2$ mm	2309.33
I-profile, $t = 1.5$ mm	1900.38
Hollow rectangular profile	725.33

The thicker I-profile ($t = 2$ mm) exhibits the largest moment of inertia and thus the greatest resistance to bending. In contrast, the hollow rectangular section performs worst in both criteria: it is heavier yet less stiff than the I-sections, making it the least efficient choice. For this reason, the hollow rectangular alternatives were discarded, reducing the number of viable configurations from 36 to 8.

The next step was to model the lightest possible configuration to verify if it satisfied the structural constraints. The configurations were evaluated sequentially from lightest to heaviest, beginning with aluminium and then moving to steel.

The imposed constraints were:

- No yielding in the structure ($\sigma_{\text{peak}} < \sigma_y$), with $\sigma_y = 230$ MPa for aluminium and $\sigma_y = 320$ MPa for steel.
- Maximum rotation at point C: $\theta_{C,\max} = 1.75° = 0.03054$ rad.
- Maximum vertical displacement at point C: $U_{2C,\max} = 2.5$ mm;

Table 5: Comparison between aluminium and steel configurations for the I-profile organised from lightest to heaviest. Bold values indicate the quantities that are within the imposed limits.

Material	Section AB	Section BC	σ_{peak} [MPa]	θ_C [rad]	U_{2C} [mm]
Aluminium	I-profile $t = 2$ mm	I-profile $t = 2$ mm	-255.8	-0.08239	-4.125
Steel	I-profile $t = 1.5$ mm	I-profile $t = 1.5$ mm	-318.5	-0.03309	**-1.660**
Steel	I-profile $t = 1.5$ mm	I-profile $t = 2$ mm	-320.2	-0.03209	**-1.618**
Steel	I-profile $t = 2$ mm	I-profile $t = 1.5$ mm	**-316.8**	**-0.02846**	**-1.417**
Steel	I-profile $t = 2$ mm	I-profile $t = 2$ mm	**-257.1**	**-0.02746**	**-1.375**

All aluminium models violate the displacement, rotation, and yielding limits. As the most rigid aluminium section ($t = 2$ mm) already fails, the remaining thinner sections, with smaller inertia and stiffness, cannot satisfy the constraints either.

The objective was to determine the lightest configuration that still satisfies all imposed constraints. Based on the analyses performed, **the steel I-profile with $t = 2$ mm in section AB and $t = 1.5$ mm in section BC** is identified as the optimal solution. This configuration ensures compliance with all displacement, rotation, and strength requirements while minimising the total mass of the structure.

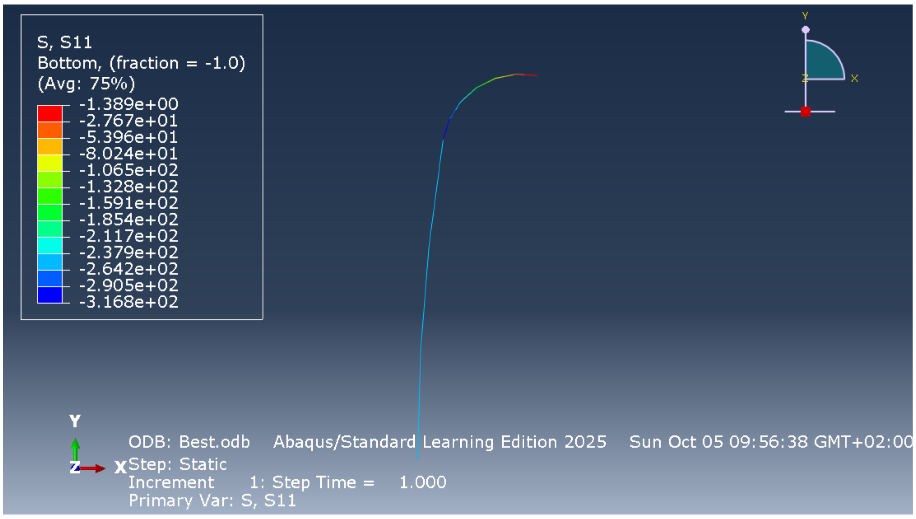

Figure 3: Normal stress S_{11} for the optimal steel I-profile design ($t_{AB} = 2$ mm, $t_{BC} = 1.5$ mm). Peak $|S_{11}| = 316.8$ MPa < 320 MPa.

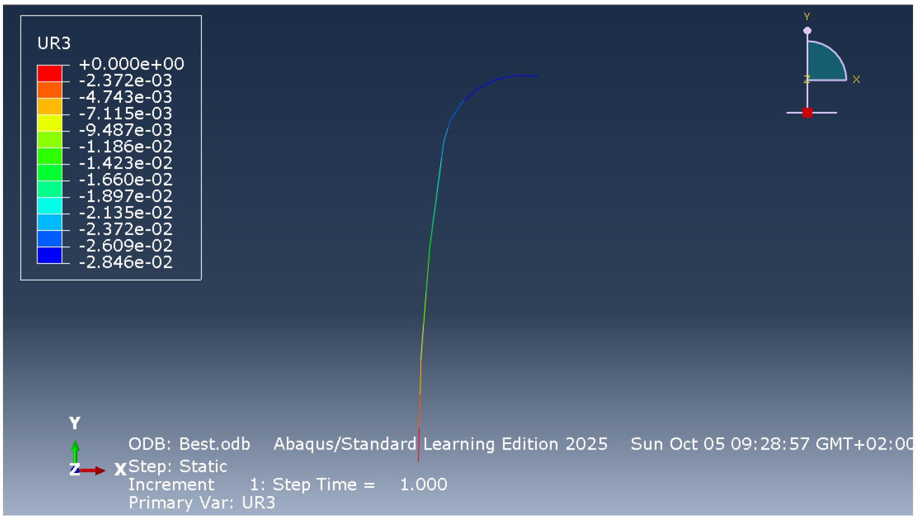

Figure 4: Rotation at point C (Abaqus variable UR3). $\theta_C = 0.02846$ rad < 0.03054 rad.

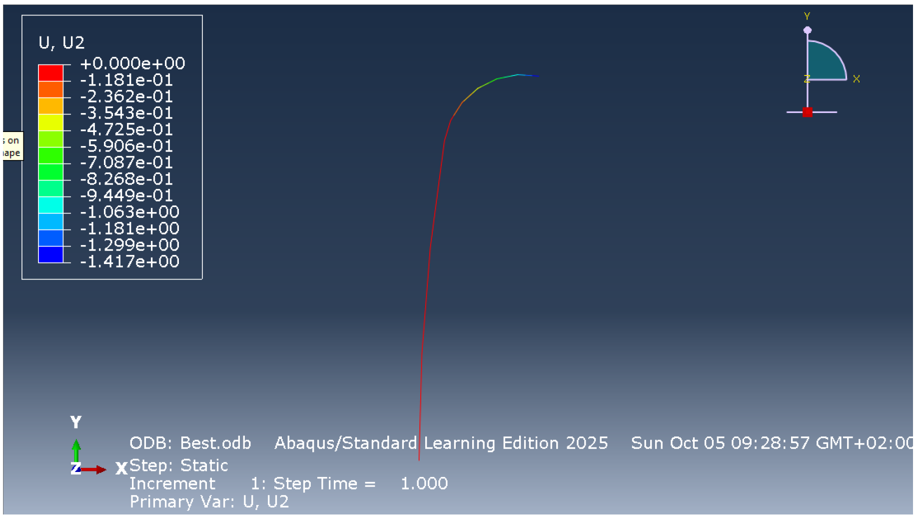

Figure 5: Vertical displacement at point C (Abaqus variable U_2). $U_{2C} = 1.417$ mm < 2.5 mm.

Reflection

As already discussed above, the preliminary geometric screening showed that among the considered shapes, the I-profiles are the most efficient in bending: for comparable masses they provide much higher flexural stiffness due to their larger second moment of area I. Consequently, the hollow rectangular option was discarded early, as it was both heavier (larger A) and less stiff (smaller I), making it an inefficient means of meeting the stiffness requirements.

Material selection then became the decisive factor. Aluminium variants could not satisfy the constraints: their lower elastic modulus E results in a smaller flexural stiffness EI for the same geometry, which leads to larger values of θ_C and U_{2C} under identical loading. Steel, on the other hand, provides sufficient stiffness and yield strength without resorting to unnecessarily large areas, thereby satisfying both serviceability and strength limits.

The distribution of thickness can be explained through the way rotation accumulates along the load path,

$$\theta_C = \int \frac{M}{EI}\, ds$$

Although the loads act primarily on member BC, the internal bending moment M increases toward the support and reaches its maximum near the base. Hence, member AB dominates the integral above and largely governs both the rotation and the vertical displacement at point C. Increasing the thickness of AB effectively reduces the ratio $\frac{M}{EI}$ where it contributes most to θ_C and U_{2C}, yielding a much larger overall stiffness improvement than adding the same mass to BC.

The stress distribution follows the same logic: the highest stresses occur close to the clamp and are relieved by stiffening AB, whereas stiffening BC alone merely transfers additional demand toward the fixed end and does not mitigate the hotspot. Therefore, the mass-optimal configuration within the given options is to retain the efficient I-profile, use steel, and concentrate thickness at the base (AB) while keeping BC as light as possible.

References

[1] D. Peeters, *Linear Modelling (incl. FEM)* , TU Delft, Version 4, 2025.